

## M003 Extension — English Worksheet (Beginner)

**Topic:** FizzBuzz Pixel Art (remainders & rules) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 3) · **Time:** about 30 minutes

You already made pixel art by choosing each block's colour **by hand**. This time the **code** chooses every colour for you — using one rule, the **FizzBuzz rule**. Load the world, type **run** in the chat, and watch the agent paint a **15 × 15** wall one block at a time, counting from **1 to 225**.

| If the block's number...  | it says          | colour   |
|---------------------------|------------------|----------|
| divides by <b>3 and 5</b> | <b>FIZZ BUZZ</b> | ● gold   |
| divides by <b>3</b>       | <b>FIZZ</b>      | ■ black  |
| divides by <b>5</b>       | <b>BUZZ</b>      | ■ purple |
| divides by neither        | <i>(nothing)</i> | ■ green  |

**Big idea:** the **number** on a block picks its colour. Divides by 3 → black. Divides by 5 → purple. Divides by 3 **and** 5 → gold. Nothing → green.

Here is one row (blocks 1 → 15). This same row repeats all the way up the wall.

🔑 **One row = blocks 1 → 15 (this row repeats up the whole wall)**

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■  | ■  | ■  | ■  | ■  | ■  |

### 1 · Predict 🧠

Use the rule. Circle the colour.

**Block 3 divides by 3. Circle its colour:** gold · black · purple · green

**Block 5 divides by 5. Circle its colour:** gold · black · purple · green

**Block 15 divides by 3 and 5. Circle its colour:** gold · black · purple · green

**Block 7 divides by nothing. Circle its colour:** gold · black · purple · green

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## 2 · Spot the Bug 🐛

**Bug** — Block 15 divides by 3 **and** 5, so it should be **gold**. In the broken code the  $\div 3$  check runs first, so block 15 turns **black** by mistake.

```
if remainder of  $N \div 3 = 0$ :  
    paint BLACK    ← FIZZ  
else if remainder of  $N \div 3 = 0$  and remainder of  $N \div 5 = 0$ :  
    paint GOLD     ← FIZZ BUZZ
```

**What colour should block 15 really be? Circle one:** gold · black

**Which check should go first? Circle one:**  $\div 3 \cdot \div 3$  and  $\div 5$

### 3 · Show Your Work 📷 🎥

Load the world. Type **run** in the chat. Watch the agent count **1** → **225** and paint the wall.

#### 🧱 The finished wall — 15 × 15 (block 1 → 225)

|    | 1     | 2     | 3     | 4     | 5      | 6     | 7     | 8     | 9     | 10     | 11    | 12    | 13    | 14    | 15   |
|----|-------|-------|-------|-------|--------|-------|-------|-------|-------|--------|-------|-------|-------|-------|------|
| 15 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 14 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 13 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 12 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 11 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 10 | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 9  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 8  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 7  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 6  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 5  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 4  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 3  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 2  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |
| 1  | green | green | black | green | purple | black | green | green | black | purple | green | black | green | green | gold |

**Every row is the same, so you see stripes going up.**

**How many gold blocks are in each row? Circle one:** 1 · 2 · 3

**Which colour is there the most of? Circle one:** gold · black · purple · green

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the colours and the stripes.

Say these out loud, filling in the blanks:

Today I built \_\_\_\_.

I built it using this code: \_\_\_\_.

In this code I used \_\_\_\_.

The hardest part was \_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

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**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.